Identification of Unknown A

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Introduction

The unknown sample was identified by analytical methods such as Nuclear Magnetic Resonance, Infrared Spectroscopy, and Mass Spectroscopy. These processes are necessary to verify the identity of a compound created through experimentation.

Methods

Nuclear Magnetic Resonance (NMR) utilizes electromagnetic radiation to reveal the structure of a compound. This form of chemical analysis is a way to understand how the carbons and hydrogens of an unknown compound are connected to one another, but will only work if the nuclei contain an odd number of either protons or neutrons (Klein 2021, p. 685). In NMR, the electromagnetic radiation creates a magnetic field which excites the nuclei in the compound. The added energy renders spin-flipping, or resonance, which is noted to calculate the structure. Signals on NMR spectrums contain the information needed to understand the compound in question. The location of the signals designate the types of protons present and how much radiation they absorb. They also reveal the integration, which is the area that is beneath each of the signals regarding the amount of that type of proton (Klein 2021, p. 685). Multiplicities, the number of peaks at a signal, indicate the amount of protons adjacent to the atom.

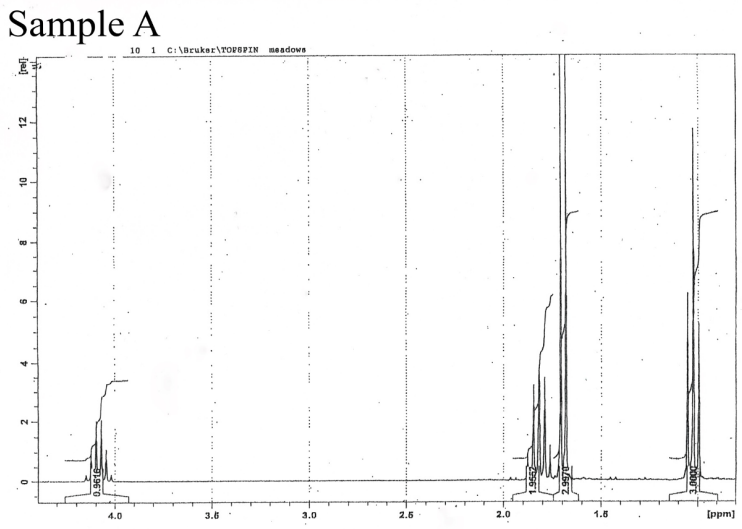
With Infrared Spectroscopy (FTIR), functional groups in compounds can be identified due to the measurable difference in vibration states created when infrared radiation is passed through a sample (Klein 2021, p. 641). Functional groups can be determined in this way because the bonds between different atoms absorb the radiation at different magnitudes, and bend or stretch depending on it. Two methods for completing IR spectroscopy are using salt plates or KBr, since they do not tamper with the IR transmittance (Klein 2021, p.641). Ranges of wave numbers indicate which functional groups are located at certain regions of the IR spectrum. All

hydrogens with a single bond to another atom fall between 4000-2700 cm^{-1}, carbons with triple bonds are between 2300-2100 cm^{-1}, carbons double-bonded to another atom are within the 1850-1600 cm^{-1} range, and all single-bonded carbons are between 1500-400 cm^{-1} (Klein 2021, p.642).

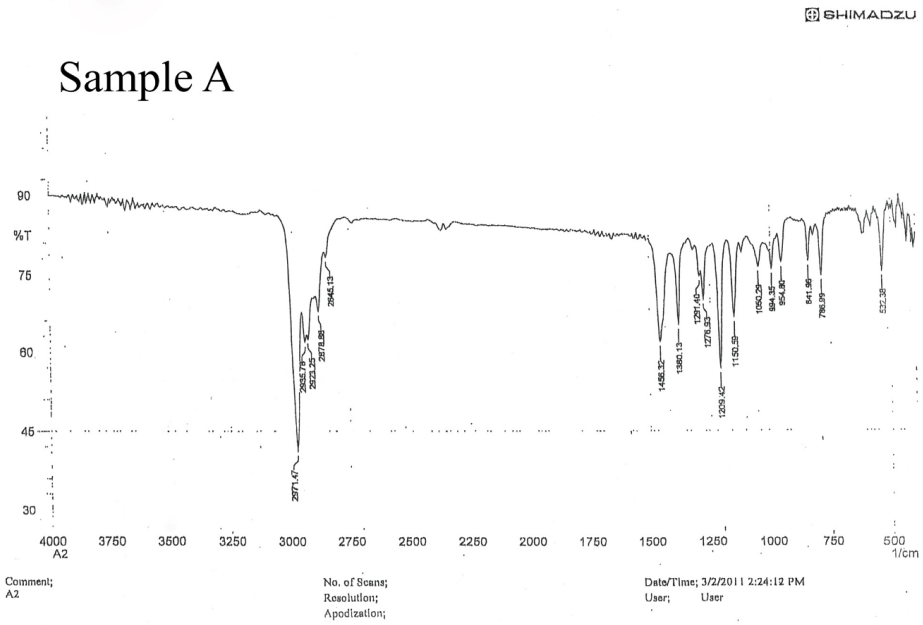
Mass spectroscopy involves ionizing the sample to discover its molecular weight and formula. The mass spectrometer vaporizes the sample and creates radical cations. It then arranges the ions with a magnetic field based on their mass-to-charge ratios (Halterman & Morvant 2013, p. 157). The ions' electricity creates a current to indicate the amount of ions with the same mass-to-charge ratio. The mass-to-charge ratio resembles the mass of the atoms, since the charge of almost all cations present is +1. The base peak is the ion that is the most abundant, found by locating the tallest peak, which usually has a relative height of 100% (Klein 2021, p. 661). The molecular ion is easily fragmented and usually splits into two strong peaks on the spectrum, however, if these are absent, the molecular ion is the same as the base peak.

Data

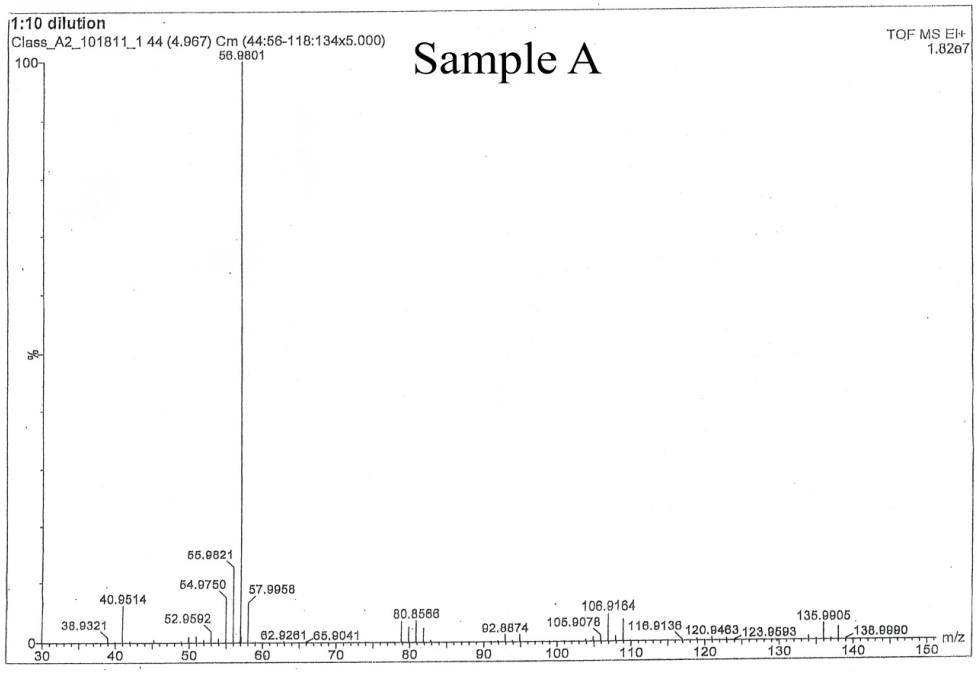
NMR Sample A



FTIR Sample A



GCMS Sample A



Analysis

The NMR spectrum integration reveals there is 1 hydrogen signal at the first peak, 2 hydrogen signals at the second, and 3 at both the third and fourth peaks, regarding a 1:2:3:3 proton ratio. The multiplicity of the first signal is a sextet, followed by a quintet, doublet, and triplet.

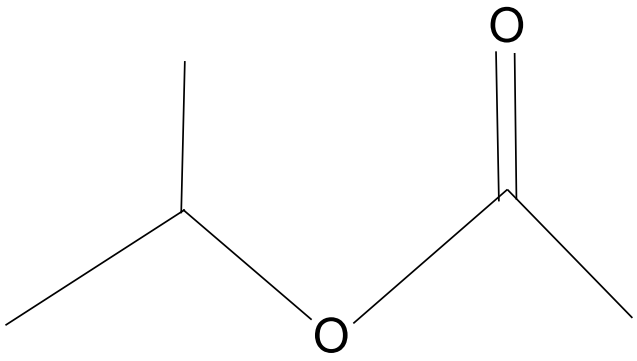
In comparison to Table 14.2, "Important Signals in IR Spectroscopy", on page 653 of *Organic Chemistry* (4th ed.), the IR spectrum for Unknown A exhibits an alkane with an sp^3 carbon at 2900 cm^{-1}, a carbon triple-bonded to a nitrogen at 2300 cm^{-1}, a phenyl group at 1450 cm^{-1}, an ester at 1200 cm^{-1}, a carbon singly-bonded to nitrogen at 1000 cm^{-1}, a carbon singly-bonded to bromine at 800 cm^{-1}, and a carbon singly-bonded to chlorine at 600 cm^{-1}.

The mass spectrum shows a base peak at 56.9801 m/z. There does not seem to be a molecular ion peak, therefore it is likely that the compound was fragmented during the process. The M+1 peak is located at 105.9078 m/z. The M+2 peak is located at 116.9136 m/z. The only strong peak in this spectrum is the base peak, and since there is not a strong M+2 peak, there are no chlorine or bromine atoms present. There are no nitrogen atoms present since there are no peaks at the molecular weight of nitrogen, which is 73. It also appears that there is not a benzene present in the compound, because they are usually "not susceptible to fragmentation, as it is the most abundant ion to pass through the spectrometer" (Klein 2021, p.661).

Conclusion

Based on the analytical methods used to determine the sample, Unknown A appears to be isopropyl acetate. While IR spectroscopy exhibited many different functional groups, the mass spectrum disproved some of them, such as the carbon-bromine, carbon-chlorine, phenyl group,

and a triple-bonded nitrogen. The mass of isopropyl acetate is close to that of the M+1 value found in mass spectrometry, which is 102.133 g/mol.



References

Halterman, R.L., & Morvant, M.C. (2013). *CHEM 2081 Organic Chemistry II Laboratory Manual* [lab manual]. Auburn (Al): Auburn University.

Klein, David R. (2021). *Organic Chemistry* (4th ed.). Wiley.